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ORIGINAL INVESTIGATION

Effect of Previous-Day Alcohol Ingestion on Muscle Function and Performance of Severe-Intensity Exercise

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Purpose: Many athletes report consuming alcohol the day before their event, which might negatively affect their performance. However, the effects of previous-day alcohol ingestion on performance are equivocal, in part, due to no standardization of alcohol dose in previous studies. The purpose of this study was to examine the impact of a standardized previous-day alcohol dose and its corresponding impact on morning-after muscular strength, muscular power, and muscular fatigue in a short-duration test and on performance of severe-intensity exercise. ***Methods:*** On 2 occasions, 12 recreationally active individuals reported to the Applied Physiology Laboratory in the evening and ingested a beverage containing either 1.09 g ethanol·kg^{-1} fat-free body mass (ALC condition) or water (PLA condition). The following morning, they completed a hangover symptom questionnaire, vertical jumps, isometric midthigh pulls, biceps curls, and a constant-power cycle ergometer test to exhaustion. The responses from ALC and PLA were compared using paired-means *t* tests. ***Results:*** Time to exhaustion in the cycle ergometer tests was less ($P = .03$) in the ALC condition (181 [39] s vs 203 [34] s; –11%, Cohen $d = 0.61$). There was no difference in performance in vertical jump test, isometric midthigh pulls, and biceps curls tests between the ALC and PLA conditions. ***Conclusions:*** Previous-day alcohol consumption significantly reduces morning-after performance of severe-intensity exercise. Practitioners should educate their athletes, especially those whose events rely on anaerobic capacity and/or a rapid response of the aerobic pathways, of the adverse effect of previous-day alcohol consumption on performance.

Keywords: ethanol, anaerobic, fatigue, power, strength

Physically active college students, including National Collegiate Athletic Association athletes, are known to ingest high volumes of alcohol.[1] Indeed, 81% of intercollegiate athletes reported alcohol consumption within the previous 12 months and 63% reported they had experienced a hangover. Green et al[2] reported that 35% of National Collegiate Athletic Association athletes participated in binge drinking during the competitive season.

Alcohol ingestion immediately before exercise might negatively affect performance (for reviews, see Vella and Cameron-Smith[3] and Barnes[4]), but most athletes do not consume alcohol immediately prior to their athletic event. O'Brien and Lyons[5] reported that 58% of athletes did consume alcohol the night before competition or practice. Whether this affects performance is equivocal.

Although previous investigations suggest that previous-day alcohol ingestion might affect morning-after exercise performance,[4] there has been limited standardization in the protocols used.[4] The purpose of this study was to investigate the effect of a standardized previous-day alcohol dose on morning-after muscular strength, power, and fatigue in a short-duration test, and performance in a constant power cycle ergometer exercise to exhaustion.

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Methods

Experimental Design

Using a within-subjects crossover experimental design, 12 university students were recruited and completed all aspects of the study. Each participant made 5 visits to the Applied Physiology Laboratory. The initial visit included anthropometric measurements, familiarization with protocols, and testing of maximal oxygen consumption (VO_2max). During the beverage ingestion visits (visits 2 and 4), participants consumed either an alcohol beverage (ALC condition) or placebo beverage (PLA condition), according to protocols used previously in our laboratory.[6–9] Participants returned the following morning for exercise testing (visits 3 and 5), which was climate-controlled and free from distractions. Performance was compared between the PLA and ALC conditions.

Participants

Six women and 6 men participated; participant characteristics are shown in Table 1. These recreationally active Kinesiology students provided written informed consent to participate. This study was approved by the Institutional Review Board of the University of North Texas and was conducted in compliance with the Declaration of Helsinki.

Potential participants were screened for medical conditions that could place them at an elevated risk during the study or could confound study results, for signs of disordered alcohol use using the Alcohol Use Disorders Identification Test[10] and specific questions on binge drinking behavior, and also for use of hormonal medications, excluding birth control medications. Participants were low to moderate alcohol consumers (≤14 standard drinks

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Table 1 Participant Characteristics, Mean (SD)

	All participants (N = 12)	Male participants (n = 6)	Female participants (n = 6)
Age, y	25.1 (4.8)	26.7 (6.4)	23.5 (2.2)
Height, cm	171.3 (7.5)	176.6 (5.9)	166.1 (4.8)
Body mass, kg	75.8 (15.5)	86.2 (14.9)	65.4 (7.0)
Body fat, %	28.2 (9.8)	26.8 (12.0)	29.6 (7.7)
Maximal oxygen consumption, $mL \cdot kg^{-1} \cdot min^{-1}$	40.7 (8.0)	38.6 (7.0)	42.7 (9.1)
Maximum heart rate, $beats \cdot min^{-1}$	181 (13)	179 (18)	182 (10)
Estimated daily energy expenditure, kJ	10,052 (1728)	11,450 (1299)	8653 (436)

per week for men and ≤7 drinks for women).[11] All participants reported familiarity with binge drinking. Eleven participants regularly consumed alcohol (at least once per month), and 8 participants routinely consumed 6 or more drinks in one sitting at least once per month.

Procedures

Visit 1: Anthropometric Measurements, Familiarization, and Establishing VO_2max. At least 2 days prior to visit 2, participants reported to the laboratory. Height and body mass were measured using a standard scale and stadiometer (seca, Chino, CA). Body composition was determined using dual-energy X-ray absorptiometry (Lunar Prodigy; General Electric, Madison, WI). Participants were familiarized with the strength and power tests and completed an exhaustive incremental exercise test on an electronically braked cycle ergometer (Lode Excalibur, Groningen, The Netherlands) using procedures previously used in our laboratory.[12] Briefly, participants were categorized into low, medium, or high fitness based on visual assessment by the primary investigator and on discussion of exercise habits with the participants. The tests began with a warm-up consisting of 4 minutes of rest followed by 3 consecutive 3-minute stages using a pedaling cadence of 80 revolutions per minute ($rev \cdot min^{-1}$). The participant's fitness informed the selection of work rates for the three 3-minute stages (low fitness: 20, 30, and 40 W; medium fitness: 30, 45, and 60 W; and high fitness: 40, 60, and 80 W). After the warm-up, the work rate was increased by a predetermined amount every 2 minutes (low fitness: 20 W; medium fitness: 30 W; and high fitness: 40 W). The highest work rate sustained for at least 60 seconds was recorded as peak power. The test was terminated when the pedaling cadence dropped below 76 $rev \cdot min^{-1}$ for 3 seconds, despite verbal encouragement. Cardiovascular–respiratory responses were monitored throughout each test using a MedGraphics UltimaO_2 metabolic cart (MedGraphics, St Paul, MN), which was calibrated according to the manufacturer's instructions.

Visits 2 and 4: Beverage Consumption Visits. The PLA and ALC conditions were scheduled with at least 3 days between visits. Consumption of the beverage commenced between 12:00 and 2:30 PM. There was no more than 2 hours' difference in beverage consumption start times between conditions for participants. Order was counterbalanced and assigned using randomization.

Participants abstained from resistance exercise, vigorous aerobic exercise, and alcohol use for at least 48 hours, and from caffeine ingestion for 12 hours, prior to these visits. They took no medications that may interfere with hangover symptoms, such as nonsteroidal anti-inflammatory drugs or antacids. Participants did not eat during the 2 hours prior to the start of each visit. Participants completed a food log of the last meal they consumed prior to the first beverage consumption visit (visit 2) and replicated that meal before visit 4. As a safety precaution, and to keep them blinded to beverage consumption condition, participants scheduled a ride home from the laboratory after each visit.

Upon arrival, participants verified compliance with all instructions. A urine sample was obtained, and urine specific gravity (USG) assessed. If USG was 1.020 or greater, participants were provided water to consume. A breathalyzer test was administered to ensure the absence of blood alcohol (Lifeloc FC10 Breathalyzer; Lifeloc Technologies, Wheat Ridge, CO).

Beverages contained either alcohol or an equivalent volume of water in a flavored, artificially sweetened, calorie-free beverage. For ALC, the dose was 1.09 g $ethanol \cdot kg^{-1}$ fat-free body mass (Vodka, 40% v/v; Smirnoff, Norwalk, CT) diluted to 15% v/v. One standard drink contains 14 g of ethanol, and participants consumed 52 (9) g of ethanol. Male participants consumed 60 (4) g of ethanol, whereas female participants consumed 44 (2) g of ethanol. Beverages were divided into 10 equal portions and 1 portion consumed each minute over a 10-minute period. The dosage averaged 0.70 g $ethanol \cdot kg^{-1}$ body mass. In both conditions, the rim of the cup was blotted with vodka to minimize the ability to distinguish the treatment condition. In ALC, peak blood alcohol concentration (BAC), estimated by breathalyzer, was 0.09 (0.02) $g \cdot dL^{-1}$ (range: 0.06–0.12 $g \cdot dL^{-1}$), and it occurred, on average, between 30 and 45 minutes after beverage consumption. The participants took, on average, between 405 and 420 minutes after consumption of the alcohol to return to a BAC of 0.00 $g \cdot dL^{-1}$. Participants were generally aware of when they were in the ALC condition due to well-known physiological and psychological side effects of alcohol intoxication but were generally unaware of when they were in the PLA condition.

In ALC, participants remained in the laboratory for at least 6.5 hours until BAC had returned to 0.00 $g \cdot dL^{-1}$. BAC was measured every 15 minutes for the first 2 hours following beverage consumption, every 60 minutes for the next 4 hours, and every 15 minutes thereafter until BAC returned to 0.00 $g \cdot dL^{-1}$. The same protocol was used for each beverage consumption visit; at no point were participants informed of their BAC or condition. In the PLA condition, participants were held for the same duration as they were in ALC (if ALC was scheduled first), or they were held for 6.5 hours (if PLA was scheduled first).

While in the laboratory, participants took one meal 3 hours after ingestion of the beverage and another 3 hours later. Each meal consisted of an amount of Papa John's™ Cheese Pizza that provided 35% of daily energy requirements, which had been

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estimated using the Harris–Benedict equation adjusted for a moderate level of activity.[13] Each meal served during visits 2 and 4 contained 3549 (589) kJ. Participants were given the option to consume 2, 1, or 0 cereal bars (Nature Valley Protein™; 765 kJ, 50% energy from fat, 29% from carbohydrates, and 21% from protein; and Nutri-Grain Soft Baked Apple & Cinnamon™; 502 kJ, 17% energy from fat, 77% from carbohydrate, and 6% from protein) during the evening after leaving the laboratory. Cereal bar consumption after visits 2 and 4 was identical. Participants consumed nothing but the cereal bars and water between beverage consumption visits and corresponding exercise testing. At the end of each beverage-consumption visit, participants were driven home, where they stayed until the next morning and then returned to the laboratory for visit 3 or 5.

Visits 3 and 5: Morning Exercise Testing. The morning following the beverage consumption visits between 8:00 and 10:00 AM (17–19 h after arrival for the corresponding beverage ingestion visit), participants returned to the laboratory and provided a urine sample. If USG was 1.020 or greater, then participants were provided water to consume until USG measured below 1.020. Participants verified compliance with all instructions before completing the Hangover Symptoms Scale questionnaire.[14] This scale requires that participants self-report the severity of 9 hangover symptoms on a scale from 0 (none) to 7 (incapacitating). Research personnel were blinded to the participant's condition.

First, participants performed 3 maximal countermovement vertical jumps with the height measured to the nearest 0.5 in (~1.2 cm) using a Vertec 2 system (Sports Imports, Hilliard, OH). After a brief rest, they performed 3 isometric mid-thigh pulls on an AccuPower System force plate (Advanced Mechanical Technology, Watertown, MA) against an immovable Olympic bar; they generated force as rapidly as possible and maintained a maximal effort for 6 seconds. Peak force against the force plate was recorded. Next, participants performed 15 repetitions of maximal biceps curls at $180°\cdot s^{-1}$ on an upper body dynamometer (Biodex Medical Systems, Shirley, NY) that was controlled with HUMAC2009 software (Computer Sports Medicine, Stoughton, MA). The mean of the measures of the average torque over the full range of motion in the first 5 repetitions was recorded as the measure of peak torque. Fatigue index was calculated as the percentage reduction in average torque from the first to final 5 repetitions.

Finally, participants completed a constant power cycle ergometer test to exhaustion. The test began with a warm-up, which consisted of 4 minutes of rest, 4 minutes of cycling at a moderate intensity (steady state) using a pedaling cadence of 80 $rev\cdot min^{-1}$, then another 4 minutes of rest. Rating of perceived exertion was obtained in the final seconds of each minute during the steady-state exercise, using the Borg 6–20 scale.[15] Following the second 4-minute rest period, participants began pedaling and quickly accelerated to 80 $rev\cdot min^{-1}$, as the work rate was increased (within 3 s) to the peak power established during visit 1. $T_{exhaustion}$ was the time from the start of the test until pedaling cadence dropped below 76 $rev\cdot min^{-1}$ for 3 seconds, despite verbal encouragement.

Measurement of VO_2 and Heart Rate. Heart rate was monitored using a Polar heart rate monitor (Polar, Lake Success, NY). Maximum heart rate (HR_{max}) was the highest 5-second value in the incremental tests, and peak heart rate (HR_{peak}) was the highest value in the constant power tests. Breath-by-breath VO_2 data were reduced to serial 15-second averages. VO_2max was the highest average of 2 adjacent 15-second values in the incremental tests, and VO_2peak was the highest such value in the constant power tests.

Calculation of Maximal Accumulated Oxygen Deficit. Breath-by-breath VO_2 data from the constant power tests were reduced to rolling 7-breath averages,[16] data from the first 20 seconds of exercise were removed,[17] and the remaining data were fit to Equation 1 using the iterative regression procedure in KaleidaGraph 4.5 software (Synergy Software, Reading, PA):

$$VO_2(t) = VO_{2baseline} + A_{primary} \times (1 - e^{-(t-TD_{primary})/\tau_{primary}}) + A_{slow} \times (1 - e^{-(t-TD_{slow})/\tau_{slow}}). \quad (1)$$

$VO_{2baseline}$ is the VO_2 at the end of the 4-minute recovery after the steady-state warm-up, $A_{primary}$ and A_{slow} are the projected increases in VO_2 due to the primary and slow component responses, $TD_{primary}$ and TD_{slow} are the time delays preceding the 2 responses, and $\tau_{primary}$ and τ_{slow} are the time constants of the 2 responses.

The area under the VO_2 response curve was determined by integration using KaleidaGraph; this accumulated VO_2 represents the aerobic contribution. The area under the primary phase was determined. The difference between these 2 areas represents the excess VO_2 attributable to the slow component. In cases in which VO_2 responses were not described adequately by 2-component model, data were fit to a monoexponential model (Equation 1, without the slow component). In order to include all participants in analyses, the effect of alcohol ingestion on VO_2 kinetics was limited to comparison of the mean response time ($TD_{primary} + \tau_{primary}$) from the monoexponential model.

Maximal accumulated oxygen deficit (MAOD), the widely accepted measure of anaerobic capacity,[18] was determined based on Medbø et al.[19] Total oxygen cost (in milliliters per kilogram) was calculated as the product of $T_{exhaustion}$ and oxygen demand plus the excess VO_2 contribution of the slow component.[20–22] MAOD was calculated by subtraction of the accumulated VO_2 from the total oxygen cost.

Statistical Analyses

After it was determined that data were normally distributed, values from the ALC and PLA were compared using paired-means *t* tests. In addition, the VO_{2peak} and HR_{peak} in each condition were compared with VO_2max and HR_{max} using a 1-way analysis of variance; as appropriate, planned post hoc comparisons of each peak value with the maximal value were performed and the results were interpreted after a Bonferroni correction. The effect size of the difference between the ALC and PLA values was calculated as Cohen *d*. All analyses were performed using SPSS (version 22; IBM, Armonk, NY). The alpha level was set at .05 and data are presented as mean (SD).

Results

$T_{exhaustion}$ in the constant power cycle ergometer tests (severe intensity) was shorter ($P = .03$) in ALC versus PLA; the effect size was medium (Cohen $d = 0.61$; Figure 1). Results of analysis of variance and post hoc comparisons revealed that VO_2max was greater than each VO_2peak value. There was no difference among HR_{max} and the two HR_{peak} values. There were no differences between PLA and ALC in responses in the steady-state warm-ups before the cycle ergometer tests. These data are shown in Table 2.

There were no differences between PLA and ALC in muscular power, isometric strength, dynamic strength, or fatigue index (Table 3).

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The number of hangover symptoms that were reported was greater in ALC (PLA, 3 [2]; ALC, 10 [5]; $P = .01$; Table 4).

Discussion

The important and novel finding in this study was that a carefully standardized previous day alcohol dose significantly reduced morning-after $T_{exhaustion}$ in severe-intensity exercise. The alcohol consumption did not affect muscular power, muscular strength, or muscular fatigue index.

$T_{exhaustion}$ in the constant power test (severe intensity) was reduced 11% in the ALC condition. This finding is consistent with the results from Karvinen et al[23] who reported that previous day alcohol consumption negatively affected morning-after severe-intensity cycling performance. In that study,[23] the alcohol dosage was quite variable (1.0–2.4 g·kg^{-1} body mass; mean ~1.67 g·kg^{-1} body mass). The test required exercising to exhaustion at a common 245 W. Significantly less work was performed in the alcohol condition. Our results are also in accordance with O'Brien[24] who found that previous day alcohol consumption reduced "aerobic performance" by 11%; the dosage was the "normal Friday night quantity of alcohol" for the 15 rugby players. Our findings are in contrast with those of Kruisselbrink et al[25] who found no reduction in running $T_{exhaustion}$ at approximately "100% maximal aerobic power" (actually, at 9.67 km·h^{-1} at 7% slope for all participants) the morning after consumption of 2, 4, or 6 bottles of beer. The authors did not provide values for $T_{exhaustion}$. Lack of standardization of the alcohol dose and lack of individualization of the exercise protocol in these studies precludes direct comparison with ours.

Factors contributing to $T_{exhaustion}$ in severe-intensity constant power exercise include VO_2max, VO_2 kinetics, anaerobic capacity, and efficiency. In our participants, VO_2peak during the constant power test was not significantly reduced (–1%, $P = .63$; effect size = 0.06) in ALC. Although not statistically significant, there was a trend with a large effect size for a greater (ie, slower) mean response time in ALC (+15%, $P = .07$; effect size = 0.96). This would require a more rapid use of the anaerobic capacity early in the exercise. However, the anaerobic capacity (MAOD) itself was not affected by the ingestion of alcohol (–3%, $P = .34$; effect size = 0.14). If anything, efficiency might be inferred to have been better in ALC, as the steady-state VO_2 in the moderate-intensity warm-ups was nonsignificantly lower in ALC (–2%, $P = .23$; effect size = 0.13). Although a lower VO_2 in ALC might suggest improved efficiency (which would be surprising), it might

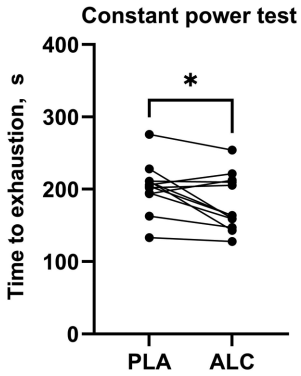

Figure 1 — Time to exhaustion during the constant-power cycle ergometer test (severe-intensity exercise domain) for participants in the PLA and ALC. Data points for each individual participant are connected by a solid line. ALC indicates alcohol condition; PLA, placebo condition. *$P < .05$.

Table 3 Muscular Power (CMJ), Isometric Strength (Peak Isometric Mid-Thigh Pull), Dynamic Strength (Biceps Curl), and Muscular Fatigue Index (Percentage Reduction in Force Production in Biceps Curl) During the Morning-After Exercise Testing in the PLA and ALC

	PLA	ALC	*t* test	Cohen *d*
CMJ, cm	50 (10)	49 (10)	$P = .08$	0.11
Isometric strength, N	2138 (490)	2161 (447)	$P = .45$	0.05
Dynamic strength, N	35 (14)	34 (13)	$P = .41$	0.05
Muscular fatigue index, %	18 (7)	20 (4)	$P = .16$	0.50

Abbreviations: ALC, alcohol condition; CMJ, countermovement vertical jump; PLA, placebo condition. Note: Data are presented as mean (SD).

Table 2 Mean (SD) Values for Responses in the Constant Power Tests During Morning-After Testing

	PLA	ALC	*t* test	Cohen *d*
Steady state (moderate-intensity exercise)				
HR, beats·min^{-1}	121 (13)	120 (12)	$P = .45$	0.11
RPE	8 (2)	8 (2)	$P = .56$	0.00
VO_2, mL·kg^{-1}·min^{-1}	13.3 (2.1)	13.0 (2.3)	$P = .23$	0.13
Respiratory exchange ratio	0.84 (0.06)	0.85 (0.06)	$P = .48$	0.18
Constant-power test (severe-intensity exercise)				
HR_{peak}, beats·min^{-1}	179 (12)	178 (11)	$P = .60$	0.09
Peak RPE	16 (2)	16 (2)	$P = .10$	0.19
VO_2peak, mL·kg^{-1}·min^{-1}	35.2 (7.3)	34.8 (7.7)	$P = .63$	0.06
$T_{exhaustion}$	203 (34)	181 (39)	$P = .03$	0.61
Mean response time, s	36 (6)	42 (7)	$P = .07$	0.96
Accumulated O_2 uptake, mL·kg^{-1}	95.8 (26.2)	80.1 (25.8)	$P = .01$	0.60
MAOD, mL·kg^{-1}	48.1 (7.7)	46.8 (10.6)	$P = .34$	0.14

Abbreviations: ALC, alcohol condition; HR, heart rate; HR_{peak}, peak HR; MAOD, maximal accumulated oxygen deficit; PLA, placebo condition; $T_{exhaustion}$, time to exhaustion during the constant-power test; RPE, rating of perceived exertion (6–20 Borg scale); VO_2, oxygen consumption; VO_{2peak}, peak VO_2.

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Table 4 Hangover Symptoms Scale

	Placebo condition		Alcohol condition	
	Number of participants experiencing symptom	Mean symptom intensity, mean (SD)	Number of participants experiencing symptom	Mean symptom intensity, mean (SD)
Hangover	0	0.0 (0.0)	7	2.4 (1.1)
Thirsty	10	1.7 (1.0)	12	2.7 (1.4)
Tired	9	2.1 (1.4)	11	3.1 (1.5)
Headache	1	1.0 (0.0)	10	2.3 (1.8)
Dizziness or faintness	0	0.0 (0.0)	0	0.0 (0.0)
Nausea	0	0.0 (0.0)	3	1.0 (0.0)
Stomachache	0	0.0 (0.0)	1	1.0 (0.0)
Heart racing	0	0.0 (0.0)	0	0.0 (0.0)
Loss of appetite	1	1.0 (0.0)	5	2.6 (1.5)

Note: Symptoms of hangover experienced before the start of morning-after testing. Mean symptom intensity is calculated among those reporting the symptom.

reflect delayed or impaired cardiorespiratory responses. Indeed, the mean response time of the VO_2 response in the warm-ups, which was calculated posteriori, was larger (42 [7] s vs 50 [7] s, +15%, $P = .04$; effect size = 1.11), meaning the response was slower, in the ALC condition. These results suggest that effects on 3 variables—VO_2max, VO_2 kinetics, and MAOD—conflated to impair performance in the ALC condition.

Although the underlying mechanisms for the reduction in $T_{exhaustion}$ remain to be elucidated, it is possible that acetaldehyde, a product of ethanol metabolism, may play a role. The toxic effects of acetaldehyde persist throughout the morning-after period even when free acetaldehyde and blood alcohol concentrations are 0. Acetaldehyde interferes with activity of glucose transporter 1[26] and has been shown, in vitro, to limit aerobic and anaerobic glucose utilization[27] through effects on signal pathways related to glucose transporter 1.[28] It has been shown, in vivo, that the effects of alcohol ingestion include decreased glycogen availability[4,5] and that a hangover is associated with reduced blood glucose concentrations, which may reduce contributions of both aerobic and anaerobic pathways.

In the present study, a standardized alcohol dose had no effect on the countermovement vertical jump measure of muscular power. This is consistent with the results of Karvinen et al[23] who found no difference in vertical jump between conditions. However, our results contrast with those of Prentice et al[29] who reported reduced jump performance the morning after alcohol ingestion. The discrepancy may be explained by differences in alcohol dose between studies; there was substantial variation in the amount of previous day alcohol consumption in Prentice et al,[29] from 6 to 10 standard drinks (2 people) to 20+ standard drinks (10 people). This is much higher than the equivalent of 3.8 standard drinks consumed, on average, in the present study. In the present study, one standard drink contained 14 g of ethanol. In addition, in Prentice et al,[29] testing was conducted over a single weekend, with baseline testing conducted on Friday and morning-after testing on Saturday. Their protocol resulted in significantly less sleep Friday night. Sleep deprivation may have contributed to the reduced muscular power that was measured on Saturday.

Our analyses revealed no difference in isometric force production in the mid-thigh pull test between conditions. This result is consistent with those of both Karvinen et al[23] and Prentice et al.[29] The similar results in those studies, and in ours, demonstrate that previous day alcohol ingestion does not affect morning-after maximal isometric strength.

We also found no effect of previous day alcohol consumption on dynamic muscular strength, which was measured in the biceps curl test. While there was not a statistically significant effect of previous day alcohol ingestion on muscular fatigue in a short-duration test, the effect size of 0.52 suggested that it was in fact compromised. This study is the first to investigate the effect of previous day alcohol ingestion on morning-after muscular fatigue index during dynamic exercise. While peak efforts—in terms of both isometric and concentric/dynamic force production and power production—were clearly not affected in the ALC condition, the results provide some evidence of a deleterious effect on muscular fatigue index. Suggestions that this effect shared a common cause with the reduced performance in the cycling exercise would be speculative.

Several studies have investigated the effect of alcohol consumption on recovery from a bout of strenuous exercise. Alcohol ingestion following muscle-damaging resistance exercise training reduces inflammatory capacity[6] and delays the recovery of strength in men[4] but not women.[7] The present study is pertinent to that situation as it focused on *performance* after alcohol consumption rather than *recovery from exercise* after alcohol consumption.

Strengths of the present study include the careful standardization of alcohol consumption and the evaluation of a range of performance measures reflecting neuromuscular, anaerobic, and aerobic qualities. A shortcoming is that, while there was no obvious difference in responses of the men and women, the study was not powered to evaluate possible interaction effects involving sex.

Practical Applications

Practitioners should educate athletes, especially those whose events rely on anaerobic capacity and/or a rapid response of the aerobic pathways, of the adverse effect of previous day alcohol consumption on performance. Future research might investigate whether participant sex, alcohol dosage, BAC, or a combination of factors, including "normal" drinking levels,[30] influences the impact of alcohol dosage on these effects.

Conclusions

Previous day alcohol consumption does not appear to affect neuromuscular qualities that influence performance in tests of muscular

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power (countermovement vertical jumps), isometric muscular strength (isometric mid-thigh pulls), and dynamic muscular strength (biceps curls), or muscular fatigue index (biceps curls). However, results from the present study demonstrate that previous day alcohol ingestion does affect morning-after severe-intensity cycling performance, and suggest that effects on MAOD, VO_2 kinetics, and VO_2max conflate and contribute to the impaired performance.

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